

KAIST

2026 Special Topics in Physics quantum Electronic Devices I:

Interferometers and Fractional Particles

Homework 6

Park Chanwoo

Department: Physics

Student ID: 20230297

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Contents

1	Kac Moody commutation relation	2
2	1D periodic tight-binding chain	3
(2a)		3
(2b)		3

Kac Moody commutation relation

Start from

$$n_{R,q} = \sum_k : c_{R,k+q}^\dagger c_{R,k} : \quad n_{R,-q} = \sum_k : c_{R,k-q}^\dagger c_{R,k} : \quad (1)$$

since the contracted term does not affect to the commutator ($[A + c, B] = [A, B] + [c, B] = [A, B]$ for any complex number c), the commutator can be calculated by:

$$[n_{R,-q}, n_{R,q}] = \sum_{k,k'} [c_{R,k-q}^\dagger c_{R,k}, c_{R,k'+q}^\dagger c_{R,k'}]. \quad (2)$$

Apply the identity

$$[AB, CD] = A\{B, C\}D - AC\{B, D\} + \{A, C\}DB - C\{A, D\}B \quad (3)$$

with $A = c_{R,k-q}^\dagger, B = c_{R,k}, C = c_{R,k'+q}^\dagger, D = c_{R,k'}$.

The anticommutators are:

$$\{B, C\} = \{c_{R,k}, c_{R,k'+q}^\dagger\} = \delta_{k,k'+q}, \quad (4)$$

$$\{B, D\} = \{c_{R,k}, c_{R,k'}\} = 0, \quad (5)$$

$$\{A, C\} = \{c_{R,k-q}^\dagger, c_{R,k'+q}^\dagger\} = 0, \quad (6)$$

$$\{A, D\} = \{c_{R,k-q}^\dagger, c_{R,k'}\} = \delta_{k-q,k'}. \quad (7)$$

Gathering the terms survive,

$$[n_{R,-q}, n_{R,q}] = \sum_{k,k'} [\delta_{k,k'+q} c_{R,k-q}^\dagger c_{R,k'} - \delta_{k-q,k'} c_{R,k'+q}^\dagger c_{R,k}] \quad (8)$$

$$= \sum_k [c_{R,k-q}^\dagger c_{R,k-q} - c_{R,k}^\dagger c_{R,k}] \quad (9)$$

To resolve $\infty - \infty$ problem, separate the operators into normal ordered part and ground state expectation value:

$$[n_{R,-q}, n_{R,q}] = \sum_k [: c_{R,k-q}^\dagger c_{R,k-q} : - : c_{R,k}^\dagger c_{R,k} :] + \sum_k [\langle c_{R,k-q}^\dagger c_{R,k-q} \rangle_0 - \langle c_{R,k}^\dagger c_{R,k} \rangle_0]. \quad (10)$$

Since the first term converges absolutely, shift the terms and compensate term-by-term. After that,

$$[n_{R,-q}, n_{R,q}] = \sum_k [\langle c_{R,k-q}^\dagger c_{R,k-q} \rangle_0 - \langle c_{R,k}^\dagger c_{R,k} \rangle_0] = \sum_k (n_{k-q}^0 - n_k^0) \quad (11)$$

Since the ground state of non-interacting fermions is fermi sea, $n_k^0 = \Theta(k_F - k)$. For large or small enough k , the term $n_{k-q}^0 - n_k^0$ is zero because both n_{k-q}^0 and n_k^0 are zero or one respectively. For simplicity, assume $q > 0$. The only non-zero contribution comes from the region near Fermi points, where $n_{k-q}^0 - n_k^0$ is one for $k - q < k_F$ and $k > k_F$. The sum is given by the number of $k \in \frac{L}{2\pi}\mathbb{Z}/L\mathbb{Z}$. Since the length of the region $k_F < k < k_F + q$ is q , the number of k in the region is $\frac{L}{2\pi}q$. Hence, we get

$$[n_{R,-q}, n_{R,q}] = \frac{qL}{2\pi}. \quad (12)$$

For $q < 0$, the region with non-zero contribution is $k_F + q < k < k_F$, same length as the previous case, so we get the same result.

1D periodic tight-binding chain

(2a)

Consider 1D periodic tight-binding chain with L sites. The Hamiltonian is given by

$$H = \epsilon_0 \sum_{i=0}^L c_i^\dagger c_i - t \sum_{i=0}^L (c_i^\dagger c_{i+1} + c_{i+1}^\dagger c_i), \quad (13)$$

the periodic boundary condition is imposed, i.e., $c_{L+1} = c_1$.

Since the system has lattice translation symmetry, it is useful to perform Fourier transformation:

$$c_k^\dagger = \frac{1}{\sqrt{L}} \sum_{j=0}^L e^{ikj} \tilde{c}_j^\dagger, \quad k \in \frac{2\pi}{L} \mathbb{Z} / L\mathbb{Z} \quad (14)$$

Explicitly, the transformation is given by

$$\sum_{i=0}^L c_i^\dagger c_i = \sum_{i=0}^L \sum_{k,k'} \frac{1}{L} e^{i(k-k')i} \tilde{c}_k^\dagger \tilde{c}_{k'} = \sum_k \tilde{c}_k^\dagger \tilde{c}_k, \quad (15)$$

$$\sum_{i=0}^L c_i^\dagger c_{i+1} = \sum_{i=0}^L \sum_{k,k'} \frac{1}{L} e^{i(k-k')i} e^{-ik'} \tilde{c}_k^\dagger \tilde{c}_{k'} = \sum_k e^{-ik} \tilde{c}_k^\dagger \tilde{c}_k. \quad (16)$$

Hence, the Hamiltonian can be written as

$$H = \sum_k (\epsilon_0 - 2t \cos k) \tilde{c}_k^\dagger \tilde{c}_k. \quad (17)$$

This gives the dispersion relation

$$E_k = \epsilon_0 - 2t \cos k. \quad (18)$$

and coefficients

$$\alpha_{kj} = \frac{1}{\sqrt{L}} e^{-ikj}. \quad (19)$$

The $k \in \frac{2\pi}{L} \mathbb{Z} / L\mathbb{Z}$ means the lattice momentum of the particle created by $\tilde{c}_k^\dagger$.

For the limit $N_e \ll L$, The dispersion relation on the region including Fermi points k_F and $-k_F$ can be approximated by quadratic dispersion relation:

$$E_k = \epsilon_0 - 2t \cos k \approx \epsilon_0 - 2t + tk^2. \quad (20)$$

(2b)

The ground state of the fermionic many-body system is given by filling the single-particle states with lowest energy. For spin-1/2 case, there are two degenerate states for each momentum k . Hence, for a five-particle system, there are four degenerate ground states, which are formed by placing two particles at $k = 0$ and distributing the remaining three particles among the four available states with $k = \pm \frac{2\pi}{L}$ and $\sigma = \uparrow, \downarrow$. Their ground state energy is given by

$$E_0 = 5\epsilon_0 - 4t - 6t \cos \frac{2\pi}{L} \approx 5\epsilon_0 - 10t + \frac{12\pi^2 t}{L^2}. \quad (21)$$

And the first excited state is $4 \times 6 = 24$ -fold degenerate, which is formed by placing two particles at $k = 0$, two particles at $k = \pm \frac{2\pi}{L}$, and the remaining particle at $k = \pm \frac{4\pi}{L}$ with $\sigma = \uparrow, \downarrow$. The energy of the first excited state is given by

$$E_1 = 5\epsilon_0 - 4t - 4t \cos \frac{2\pi}{L} - 2t \cos \frac{4\pi}{L} \approx 5\epsilon_0 - 10t + \frac{24\pi^2 t}{L^2}. \quad (22)$$